



SIMATS ENGINEERING

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Course Code: CSA0303

Slot: C

Course Name: DATA STRUCTURES

Course Faculty: Dr. Rajesh Kumar

Project Title: Design and implementation of a Web Browser with Stack Data Structures.

Module Photographs:



Project Description:

The project applies the **Stack data structure** to model essential web browser operations such as back navigation, forward navigation, undo, and redo. The system uses the **Last-In-First-Out (LIFO)** principle to efficiently manage recently visited webpages and text-editing actions. Separate stacks are used for **Back, Forward, Undo, and Redo operations** to maintain independent histories and ensure correct navigation and editing behavior. When a user navigates backward and then opens a completely new webpage, the existing Forward Stack is cleared to establish a new browsing path. Stack operations such as **Push, Pop, and Clear** provide an efficient way to manage these actions. The design also analyzes why a single stack is insufficient for handling all browser functionalities simultaneously. The project demonstrates the practical application of stacks in a real-world system while highlighting their simplicity, efficiency, and **$O(1)$ time complexity** for basic insertion and deletion operations.

Student Signature

Guide Signature